

Devaney-chaotic Rolewicz sums are densely distributionally chaotic

Candidate full solution to the conjecture in arXiv:1504.02445

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Status. Candidate full solution, pending expert review.

Source. D. Bongiorno, U. B. Darji, and L. Di Piazza, *Rolewicz-type chaotic operators* [1]. The conjecture is on printed page 4 of the source PDF (arXiv:1504.02445).

1 The question

For a strictly increasing map $f : \mathbb{N} \rightarrow \mathbb{N}$, define

$$T_f(x_1, x_2, \dots) = (x_{f(1)}, x_{f(2)}, \dots) \quad (x \in \ell_p).$$

Two such maps f, g are *almost disjoint* in the source’s sense if $\{k : f(k) = g(k)\}$ is finite and $f(k) \neq g(k')$ whenever $k \neq k'$. The source conjectures that, for pairwise almost-disjoint $f_1, \dots, f_t$ and real $c_1, \dots, c_t$, Devaney chaos of

$$T = \sum_{i=1}^t c_i T_{f_i} : \ell_p \longrightarrow \ell_p \tag{1}$$

implies distributional chaos.

Transcription. “We conjecture that if $f_1, \dots, f_t$ are strictly increasing pairwise almost disjoint functions and $c_1, \dots, c_t$ are real numbers, then $T = \sum_{i=1}^t c_i T_{f_i}$ being chaotic (in the sense of Devaney) implies that T is distributionally chaotic.”

2 Proof intuition

Far enough along the coordinates, almost disjointness turns T into a perfect branching backward shift: every input coordinate is used in at most one branch and row. The norm of a tail row is therefore exactly the dual norm

$$C = \|(c_1, \dots, c_t)\|_{p'}.$$

Hypercyclicity passes to this tail factor and forces $C > 1$.

There may be collisions in finitely many initial rows. Hypercyclicity forces each such row to have a nonzero coefficient that eventually escapes into the collision-free tail; otherwise the adjoint would have a nonzero finite-dimensional invariant subspace. Beyond escape, coefficients branch with dual norm growing like C^N . Norming functionals therefore produce a right inverse R_N of T^N with $\|R_N\| \leq KC^{-N} < 1$.

is chaotic. The precise formulation is given in Theorem 14. As corollaries, we obtain the following results which show that set of chaotic operators contains large algebraic structure.

Corollary 11 *There exists an infinite family $\mathcal{T}$ of chaotic operators such that for all $T_1, T_2, \dots, T_t \in \mathcal{T}$ and $c_1, c_2, \dots, c_t \in \mathbb{R}$ the operator $\lambda \sum_{i=1}^t c_i T_{f_i}$ is chaotic for sufficiently large λ , provided that the following sum $\sum_{i=1}^t c_i T_{f_i}$ is not the zero operator.*

Corollary 14 *There exists a family $\mathcal{T}$ of cardinality continuum of chaotic operators such that for almost all $(c_1, c_2, \dots, c_t) \in \mathbb{R}^t$ and $T_1, T_2, \dots, T_t \in \mathcal{T}$ the operator $\lambda \sum_{i=1}^t c_i T_{f_i}$ is chaotic for sufficiently large λ .*

We would like to point out that recently other notions of chaos have enjoyed attention in linear dynamics as well. These notions include Li-Yorke chaos and distributional chaos. A characterization of Li-Yorke chaos in Banach spaces in terms of irregular vectors was given in [5] and later extended to the setting of Fréchet spaces in [8]. As an easy consequence of these results one obtains that all hypercyclic operators are Li-Yorke chaotic. Hence, all operators constructed in our main results are Li-Yorke chaotic. A characterization of distributional chaos in the setting of Fréchet spaces in terms of distributionally irregular vectors was given in [9]. We conjecture that all operators constructed in our main results are distributionally chaotic. Indeed, we conjecture that if $f_1, \dots, f_t$ are strictly increasing pairwise almost disjoint functions and $c_1, \dots, c_t$ are real numbers, then $T = \sum_{i=1}^t c_i T_{f_i}$ being chaotic (in the sense of Devaney) implies that T is distributionally chaotic. Indeed, for weighted backward shifts it was shown in [20] that Devaney chaos implies distributional chaos. We conjecture that using the characterization of distributional chaos given in [9], one may be able to prove a large class of Devaney chaotic operators, which include weighted shifts as well as Rolewicz-type operators defined in this article, are distributionally chaotic.

1. Notations and preliminaries

In what follows Y is a separable infinite dimensional Banach space. Let

Figure 1: The conjecture in the original paper, printed page 4. This is a full-width crop rendered from the archived source PDF.

A contracting right inverse gives shadowing by a convergent correction series. Thus T has finite shadowing. A theorem of Bernardes and Peris [2, Theorem 11] then upgrades Devaney chaos plus finite shadowing to *dense* distributional chaos. The latter is stronger than the conjectured conclusion.

3 The theorem

Let p' denote the conjugate exponent when $p > 1$, and put $p' = \infty$ when $p = 1$.

Theorem 1. *Let $1 \leq p < \infty$, let $f_1, \dots, f_t : \mathbb{N} \rightarrow \mathbb{N}$ be strictly increasing and pairwise almost disjoint, and let T be given by (1). If T is hypercyclic, then some power T^N has a bounded right inverse R_N satisfying $\|R_N\| < 1$. Consequently T has the positive and finite shadowing properties.*

If T is Devaney chaotic, then T is densely distributionally chaotic. In particular, the conjecture of Bongiorno–Darji–Di Piazza is true.

We split the proof into four elementary lemmas.

Lemma 1 (collision-free tail). *There is $m \in \mathbb{N}$ such that, on $Y = \overline{\text{span}}\{e_k : k \geq m\}$, the factor D determined by*

$$P_Y T = D P_Y$$

has norm

$$C := \|D\| = \|(c_1, \dots, c_t)\|_{p'}. \quad (2)$$

If T is hypercyclic, then $C > 1$.

Proof. Choose m so that $f_i(k) \neq f_j(k)$ for $i \neq j$ and $k \geq m$. Strict increase gives $f_i(k) \geq k$, so coordinates $k \geq m$ depend only on the tail and $P_Y T = D P_Y$. For $k \geq m$, all $f_i(k)$ are distinct. Moreover, if $k \neq \ell$, almost disjointness gives $f_i(k) \neq f_j(\ell)$ for every i, j . Hence all input coordinates appearing in the tail rows are globally distinct.

Hölder's inequality, followed by this disjointness, gives

$$\|Dy\|_p^p = \sum_{k \geq m} \left| \sum_i c_i y_{f_i(k)} \right|^p \leq C^p \sum_{k \geq m} \sum_i |y_{f_i(k)}|^p \leq C^p \|y\|_p^p.$$

Equality is attained by putting a dual norming vector for (c_i) on the children of one row. The same proof with the maximum norm covers $p = 1$. Thus (2) holds.

If x is hypercyclic for T , then $P_Y x$ is hypercyclic for D , since $P_Y T^n x = D^n P_Y x$ and the projection of a dense orbit is dense. An operator of norm at most one has a bounded orbit from every vector and cannot be hypercyclic on the unbounded space Y . Therefore $C > 1$. $\square$

For a word $\sigma = (i_1, \dots, i_r)$, write $f_\sigma = f_{i_r} \circ \dots \circ f_{i_1}$ and $c_\sigma = c_{i_1} \dots c_{i_r}$.

Lemma 2 (finite-prefix escape). *Assume that T is hypercyclic. For every $k < m$ there are $r_k \geq 0$, an index $a_k \geq m$, and a nonzero scalar β_k such that β_k is the coefficient of $e_{a_k}^*$ in $(T^*)^{r_k} e_k^*$.*

Proof. If no such coefficient existed, the cyclic subspace

$$M_k = \text{span}\{(T^*)^n e_k^* : n \geq 0\}$$

would be a nonzero T^* -invariant subspace of $\text{span}\{e_1^*, \dots, e_{m-1}^*\}$, hence finite-dimensional. Its annihilator would give a nonzero finite-dimensional factor of T . A dense orbit of T would project to a dense orbit on that factor, contradicting the standard fact that no linear operator on a nonzero finite-dimensional real or complex space is hypercyclic. $\square$

For $k \geq m$, set $r_k = 0$, $a_k = k$, and $\beta_k = 1$.

Lemma 3 (contracting right inverse of a power). *If T is hypercyclic, there are $K < \infty$ and, for every sufficiently large N , a bounded operator R_N such that*

$$T^N R_N = I, \quad \|R_N\| \leq K C^{-N}. \quad (3)$$

In particular, $\|R_N\| < 1$ for some N .

Proof. Fix $N \geq \max_{k < m} r_k$. Once the path from k reaches $a_k \geq m$, distinct continuation words never collide. Indeed, equality of two equal-depth descendants can be pulled back one level using injectivity and the cross-input clause in almost disjointness; at a common input at least m , the choice of branch is also unique. Likewise, equal-depth descendant sets from distinct starting rows are disjoint.

It follows that the coordinates

$$J_{k,N} = \{f_\sigma(a_k) : |\sigma| = N - r_k\}$$

occur in row k of T^N with coefficients $(\beta_k c_\sigma)_\sigma$, and occur in no other row. Their $\ell_{p'}$ norm is

$$|\beta_k| \left(\sum_{|\sigma|=N-r_k} |c_\sigma|^{p'} \right)^{1/p'} = |\beta_k| C^{N-r_k},$$

with the evident maximum-norm formula for $p = 1$. By ℓ_p - $\ell_{p'}$ duality there is a vector $u_{k,N}$ supported on $J_{k,N}$ such that

$$T^N u_{k,N} = e_k, \quad \|u_{k,N}\|_p = |\beta_k|^{-1} C^{-(N-r_k)}. \quad (4)$$

The supports of the vectors $u_{k,N}$ are pairwise disjoint. Thus the rule $R_N e_k = u_{k,N}$ extends boundedly to ℓ_p , satisfies $T^N R_N = I$, and has norm equal to the supremum of the column norms in (4). Since only the indices $k < m$ have exceptional (r_k, β_k) , (3) holds with

$$K = \max \left\{ 1, \max_{k < m} |\beta_k|^{-1} C^{r_k} \right\}.$$

Now $C > 1$ by Lemma 1, so $K C^{-N} < 1$ for large N . □

Lemma 4 (right inverse implies shadowing). *If a bounded operator A on a Banach space has a right inverse R with $\|R\| = \rho < 1$, then A has positive shadowing. If T^N has positive shadowing, then T has positive shadowing.*

Proof. Let $(x_j)_{j \geq 0}$ be a positive pseudo-orbit for A and write $e_j = x_{j+1} - A x_j$. The series

$$y = x_0 + \sum_{j=0}^{\infty} R^{j+1} e_j \quad (5)$$

converges. Since $AR = I$, cancellation of the first n error terms gives

$$A^n y - x_n = \sum_{j=n}^{\infty} R^{j+1-n} e_j, \quad \|A^n y - x_n\| \leq \frac{\rho}{1-\rho} \sup_j \|e_j\|.$$

This proves positive shadowing for A .

For the second assertion, group a δ -pseudo-orbit of T into blocks of length N . Its sampled sequence at times $0, N, 2N, \dots$ is a $\delta \sum_{r=0}^{N-1} \|T\|^r$ -pseudo-orbit for T^N . Shadow the sampled sequence, then propagate over each of the finitely many intermediate times. The estimate

$$\|T^r(T^{jN} y - x_{jN})\| + \delta \sum_{s=0}^{r-1} \|T\|^s, \quad 0 \leq r < N,$$

is made arbitrarily small by first choosing the shadowing tolerance for T^N and then δ . Hence T has positive shadowing. □

Proof of Theorem 1. Lemmas 1–4 show that a hypercyclic T has positive shadowing. Positive shadowing immediately implies finite shadowing. If T is Devaney chaotic, Bernardes and Peris [2, Theorem 11] apply on the Fréchet space ℓ_p and imply that T is densely distributionally chaotic. This proves, and strengthens, the source conjecture. □

4 Verification and limitations

The exact-arithmetic script `code/verify_right_inverse.py` checks the dual normalization, disjoint supports, and $T^N R_N = I$ for a finite two-branch tree with rational coefficients. It is only a sanity check; the proof above is symbolic.

The theorem does not assert that every formal sum (1) is chaotic; it starts from the conjecture’s exact conditional hypothesis. In particular, it does not rely on the source paper’s separate sufficient condition for chaos. The proof also genuinely uses the source’s strong cross-input almost-disjointness condition, not merely eventual pointwise inequality of the functions.

Novelty check. On 12 August 2026, bounded searches of arXiv and the run indexes used the exact title and arXiv id together with “distributionally chaotic”, “shadowing”, “finite shadowing”, “pseudo-shift”, and “generalized backward shift”. They found the general Bernardes–Peris theorem but no paper explicitly resolving this conjecture or giving the contracting-power right-inverse argument for these almost-disjoint sums. Novelty confidence is therefore moderate, not definitive.

Human-review recommendation. Prioritize review. The central points to check are the escape lemma and the absence of descendant collisions used in Lemma 3; after those, the quantitative contraction and shadowing steps are routine.

References

- [1] D. Bongiorno, U. B. Darji, and L. Di Piazza, *Rolewicz-type chaotic operators*, arXiv:1504.02445 (2015).
- [2] N. C. Bernardes Jr. and A. Peris, *On shadowing and chain recurrence in linear dynamics*, arXiv:2305.02714 (2023), Theorem 11.